

# Entropy factorization on every $\ell_p$ sphere

Candidate full solution to Conjecture 1 in arXiv:2407.13457

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**Status: candidate full result; likely valid; human review needed.** The conjectured entropy factorization holds for every  $p > 0$  and every block weight vector. In fact, the same theorem holds for every symmetrized Dirichlet law with arbitrary positive parameters.

## Source conjecture

Caputo and Salez ask whether cone measure on every  $\ell_p$  sphere satisfies the same arbitrary-block entropy factorization known for the Euclidean sphere [1, Conjecture 1, p. 16]. Their 2026 journal publication retains this conjecture. The exact arXiv source text is reproduced below.

where  $G = (G_1, \dots, G_n)$  is a vector of i.i.d. real random variables with density proportional to  $e^{-|x|^p}$ ,  $x \in \mathbb{R}$ .  $\mathbf{P}$  is also called the *cone measure* on  $\mathbb{S}_p^{n-1}$ ; see e.g. [7]. A characteristic property of this construction is that  $X$  and  $S_p(G)$  are independent. Note that, for  $p = 1, 2$ ,  $\mathbf{P}$  coincides with the uniform distribution on  $\mathbb{S}_1^{n-1}$ , and  $\mathbb{S}_2^{n-1} = \mathbb{S}^{n-1}$  respectively. We believe that it is possible to prove the following statement as a consequence of Theorem 2.

**Conjecture 1.** *For all  $p > 0$ , all probability vectors  $\theta$ , the cone measure  $\mathbf{P}$  on  $\mathbb{S}_p^{n-1}$  satisfies*

$$\forall f \in L \log L, \quad \theta_{**} \text{Ent}(f) \leq \sum_{A \subset [n]} \theta_A \mathbf{E}[\text{Ent}_A(f)]. \quad (39)$$

For  $p = 2$  this coincides with the result in [6, Proposition 11]. A related estimate for all  $p > 0$ , under additional symmetry assumptions on  $f$ , was obtained in [6, Corollary 16].

For a probability measure  $\nu$  and a nonnegative  $F \in L \log L(\nu)$ , write

$$\text{Ent}_\nu(F) = \int F \log F \, d\nu - \left( \int F \, d\nu \right) \log \left( \int F \, d\nu \right).$$

If the variables are indexed by  $[n]$ , let  $\text{Ent}_A(F)$  denote entropy in the coordinates in  $A$ , conditionally on all coordinates in  $A^c$ . For a probability vector  $\theta = (\theta_A)_{A \subset [n]}$ , put

$$\theta_{**} := \min_{1 \leq i < j \leq n} \sum_{A \supset \{i, j\}} \theta_A, \quad \theta_* := \min_{1 \leq i \leq n} \sum_{A \ni i} \theta_A.$$

## Proof intuition

Under cone measure, the signs of the coordinates are independent fair signs, independent of the magnitudes

$$Z_i = |X_i|^p, \quad Z \sim \text{Dirichlet}(1/p, \dots, 1/p).$$

The magnitude layer already has the desired  $\theta_{**}$  factorization: it is a direct entropy-dual form of the Dirichlet Brascamp–Lieb inequality in [2, Proposition 14, p. 13]. The sign layer has ordinary product Shearer factorization with the larger constant  $\theta_* \geq \theta_{**}$ .

The only issue is that a block update resamples signs and magnitudes together. The entropy chain rule splits each joint block entropy into exactly those two pieces. Convexity of conditional entropy shows that the magnitude term needed after averaging all signs is no larger than the magnitude term actually present in the joint block update. Adding the two estimates proves the conjecture, without constructing any geometric regularizer.

## A Dirichlet factorization lemma

Let

$$\Delta_{n-1} = \{z \in [0, 1]^n : \sum_i z_i = 1\}$$

and let  $\mu_\alpha$  be the Dirichlet law with arbitrary parameters  $\alpha = (\alpha_1, \dots, \alpha_n) \in (0, \infty)^n$ .

**Lemma 1.** *For every nonnegative  $h \in L \log L(\mu_\alpha)$ ,*

$$\theta_{**} \text{Ent}_{\mu_\alpha}(h) \leq \sum_{A \subset [n]} \theta_A \mathbf{E}_{\mu_\alpha}[\text{Ent}_A(h)]. \quad (1)$$

*Proof.* If  $\theta_{**} = 1$ , every set with positive weight contains every pair and hence equals  $[n]$ , so the claim is immediate. Suppose  $\theta_{**} < 1$  and set  $c_A = \theta_A / (1 - \theta_{**})$ .

Apply the Dirichlet Brascamp–Lieb inequality [2, Proposition 14] to functions  $g_A$  depending on the coordinates in  $A^c$ . Its coefficient condition holds because, for every  $i \neq j$ ,

$$\sum_{A: A^c \cap \{i, j\} \neq \emptyset} c_A = \frac{1 - \sum_{A \supset \{i, j\}} \theta_A}{1 - \theta_{**}} \leq 1.$$

(When  $A = [n]$ ,  $g_A$  is constant and its factor is harmless.) Thus

$$\int \prod_A g_A^{c_A} d\mu_\alpha \leq \prod_A \left( \int g_A d\mu_\alpha \right)^{c_A}.$$

The standard entropy–Brascamp–Lieb duality, recalled as Theorem 1 in the source paper [1], converts this into

$$\sum_A \theta_A \text{Ent}_{\mu_\alpha}(\mathbf{E}_A h) \leq (1 - \theta_{**}) \text{Ent}_{\mu_\alpha}(h).$$

Finally, the conditional entropy chain rule

$$\text{Ent}(h) = \text{Ent}(\mathbf{E}_A h) + \mathbf{E}[\text{Ent}_A(h)]$$

and  $\sum_A \theta_A = 1$  give (1). □

## The symmetrized Dirichlet theorem

Let  $\sigma_n$  be uniform measure on  $\{-1, 1\}^n$ , independent of  $Z \sim \mu_\alpha$ . Equivalently,  $(Z, \varepsilon)$  induces on

$$\Sigma_{n-1}^\pm = \{y \in \mathbb{R}^n : \sum_i |y_i| = 1\}$$

the symmetrized Dirichlet law via  $y_i = \varepsilon_i Z_i$ . This is precisely the class for which [2, Remark 16, p. 14] identified a sign/ergodicity obstruction.

**Theorem 1.** For arbitrary  $\alpha_i > 0$ , every probability vector  $\theta$ , and every nonnegative  $F \in L \log L(\mu_\alpha \otimes \sigma_n)$ ,

$$\theta_{**} \text{Ent}(F) \leq \sum_{A \subset [n]} \theta_A \mathbf{E}[\text{Ent}_A(F)]. \quad (2)$$

*Proof.* Write  $\bar{F}(z) = \mathbf{E}_\varepsilon F(z, \varepsilon)$ . The global entropy chain rule gives

$$\text{Ent}(F) = \text{Ent}_Z(\bar{F}) + \mathbf{E}_Z \text{Ent}_\varepsilon(F). \quad (3)$$

Fix a block  $A$ . Given  $(Z_{A^c}, \varepsilon_{A^c})$ , the block variables consist of the conditional Dirichlet vector  $Z_A$  and independent fair signs  $\varepsilon_A$ . Applying the chain rule inside this conditional law gives

$$\mathbf{E} \text{Ent}_A(F) = M_A + S_A, \quad (4)$$

where

$$\begin{aligned} M_A &= \mathbf{E}_{Z_{A^c}, \varepsilon_{A^c}} \text{Ent}_{Z_A|Z_{A^c}}(\mathbf{E}_{\varepsilon_A} F), \\ S_A &= \mathbf{E}_Z \mathbf{E}_{\varepsilon_{A^c}} \text{Ent}_{\varepsilon_A}(F). \end{aligned}$$

Lemma 1, applied to  $\bar{F}$ , controls the magnitude part. For each fixed  $Z_{A^c}$ , conditional entropy is a convex functional of its nonnegative argument. (Its second variation in direction  $u$  is  $\mathbf{E}[u^2/h] - (\mathbf{E}u)^2/\mathbf{E}h \geq 0$ .) Since

$$\bar{F} = \mathbf{E}_{\varepsilon_{A^c}} [\mathbf{E}_{\varepsilon_A} F],$$

this convexity and Lemma 1 yield

$$\theta_{**} \text{Ent}_Z(\bar{F}) \leq \sum_A \theta_A M_A. \quad (5)$$

For every fixed  $z$ , ordinary Shearer's inequality on the product sign space gives

$$\theta_* \text{Ent}_\varepsilon(F(z, \cdot)) \leq \sum_A \theta_A \mathbf{E}_{\varepsilon_{A^c}} \text{Ent}_{\varepsilon_A}(F(z, \cdot)).$$

Moreover  $\theta_{**} \leq \theta_*$ : choose an index attaining the minimum in  $\theta_*$  and pair it with any other index. Integrating in  $z$  therefore gives

$$\theta_{**} \mathbf{E}_Z \text{Ent}_\varepsilon(F) \leq \sum_A \theta_A S_A. \quad (6)$$

Adding (5) and (6), then using (3)–(4), proves (2). The argument first applies to bounded positive  $F$ ; standard  $L \log L$  truncation gives the stated form.  $\square$

## Resolution of the source conjecture

**Corollary 1.** For every  $n \geq 2$ ,  $p > 0$ , probability vector  $\theta$ , and cone measure  $\mathbb{P}$  on the  $\ell_p$  sphere  $\mathbb{S}_p^{n-1}$ ,

$$\theta_{**} \text{Ent}_{\mathbb{P}}(f) \leq \sum_{A \subset [n]} \theta_A \mathbf{E}_{\mathbb{P}}[\text{Ent}_A(f)] \quad (f \in L \log L).$$

Thus Conjecture 1 in [1] is true.

*Proof.* Under cone measure,

$$Z_i = |X_i|^p \sim \text{Dirichlet}(1/p, \dots, 1/p), \quad \varepsilon_i = \text{sgn}(X_i)$$

are independent, with the signs fair and independent. The coordinatewise map  $(z, \varepsilon) \mapsto (\varepsilon_i z_i^{1/p})_{i=1}^n$  is an almost-sure bijection onto  $\mathbb{S}_p^{n-1}$  and preserves every complementary-coordinate  $\sigma$ -field. Conditional block entropies are therefore unchanged, and Theorem 1 with  $\alpha_i = 1/p$  applies.  $\square$

## Scope, verification, and novelty

The proof covers all  $p > 0$ , including  $p < 1$ , because it uses only Dirichlet laws with positive parameters; no norm convexity or boundary regularity is invoked. Dirichlet coordinates are positive almost surely, so coordinate signs and complementary  $\sigma$ -fields are unambiguous modulo null sets. The theorem proves the conjectured constant but does not assert that it is optimal for every choice of  $p$  and  $\theta$ .

The run’s four cheap indexes contained no entry for arXiv:2407.13457. Targeted searches through 13 August 2026 used the exact conjecture phrase, “symmetrized Dirichlet”, “cone measure”, and arbitrary-block  $\ell_p$ -sphere entropy factorization. They found the source, its 2026 journal publication (which retains the conjecture), and the cited 2011 Dirichlet result, but no later primary-source resolution. Novelty remains provisional pending expert literature review.

The accompanying deterministic Gauss–Jacobi script tests fifteen genuinely sign-dependent functions for  $n = 3$  and Dirichlet parameters 0.2, 0.5, 1, 2, 5; all pass. This is supporting QA, not part of the proof. The recommended human-review focus is the one-line Brascamp–Lieb coefficient calculation in Lemma 1 and the convexity comparison leading to (5).

## References

- [1] P. Caputo and J. Salez, *Entropy factorization via curvature*, J. Funct. Anal. 291 (2026), 111498; arXiv:2407.13457.
- [2] F. Barthe, D. Cordero-Erausquin, M. Ledoux, and B. Maurey, *Correlation and Brascamp–Lieb inequalities for Markov semigroups*, Int. Math. Res. Not. 2011 (2011), 2177–2216; arXiv:0907.2858.